FFGFT — Falsifiability

Physical Theory vs. Metaphysical Speculation

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Companion document to Dok. 248 (Epistemological Position)

https://github.com/jpascher/T0-Time-Mass-Duality

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Abstract

This document is a companion to Dok. 248 (FFGFT — Epistemological Position). It clarifies the philosophy-of-science position of FFGFT against metaphysical speculation: FFGFT is a physical theory with empirical content and concrete falsification conditions. The postulate of "pure, unrestrained information" as a primitive (Myser 2026) is, by contrast, a metaphysical framing without derivation power — not falsifiable, not connectable to known physics. The decisive difference lies not only in the choice of primitive, but in the consequences of that choice.

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.1 Falsifiability as a Scientific Criterion

A scientific theory distinguishes itself from metaphysical speculation through its **vulnerability to refutation** [?]. A theory that cannot be falsified by any conceivable experiment makes no physical claims — it makes claims about what we consider possible, not about what the world is.

This criterion is not a dogma but a methodological tool: it draws a clear line between theories that expose themselves to the world and those that evade it.

Popper's Criterion

A theory is scientific if it is **principally falsifiable**: if there exist observations or experiments that would *contradict* the theory and thereby refute it. A theory compatible with every possible observation says nothing about the world.

.2 FFGFT Is Falsifiable

FFGFT makes concrete, predictive claims that could in principle be refuted.

Minimal Length Scale L_0

$$L_0 = \xi \cdot \ell_P \approx 2.2 \times 10^{-39} \text{ m} \tag{1}$$

This absolute lower bound on spacetime structure is a prediction of FFGFT. If future high-energy experiments or cosmological observations consistently showed structures *below* this scale, FFGFT would be falsified. Conversely, if structures consistently remained at or above this bound, this supports the prediction.

Fractal Dimension D_f

The fractal dimension of spacetime in FFGFT is:

$$D_f = 2.999867 \tag{2}$$

This is a deviation from exactly 3 by a factor of 1.33×10^{-4}. This deviation is in principle measurable via:

- **Gamma-ray burst time-of-flight dispersion**: Energy-dependent light travel time from cosmological sources is an established search channel for quantum gravity effects. FFGFT predicts a specific, ξ-dependent dispersion structure.
- **Gravitational wave phase structure**: Deviations in wave dispersion at extremely high frequencies.

If a measurement showed a deviation incompatible with $D_f = 2.999867$, FFGFT would be falsified.

No Singularity in Black Holes

FFGFT predicts that black holes contain no singularity — because L_0 sets an absolute lower density bound that cannot be structurally violated.

This prediction is distinguishable from general relativity through:

- **Gravitational wave echoes**: Quantum gravity models without singularities predict characteristic echo signals after mergers, potentially detectable by future generations of LIGO/Virgo.
- **Hawking radiation spectrum**: FFGFT predicts a modified spectrum reflecting the T^4 winding structure — distinguishable from the purely thermal spectrum of standard theory.

Coupling Constants from ξ

FFGFT derives all fundamental coupling constants from a single parameter $\xi = 1.333 \times 10^{-4}$. Every experimental refinement of these constants is a test of FFGFT:

- Fine structure constant $\alpha^{-1} = 137.036$ — FFGFT derivation: $\alpha^{-1} \approx 137.036$ (deviation $< 0.001\%$)
- Anomalous magnetic moment of the electron a_e — FFGFT correction: $K_{\text{frak}}^{3/2}$ reduces deviation from 2.03% to 0.014%
- Koide formula for lepton masses — FFGFT provides a geometric derivation

Each of these derivations can be confirmed or refuted by more precise measurements.

.3 The Postulate of "Pure Information" Is Not Falsifiable

Atticus Myser's postulate — "pure unrestrained information" as primitive, without structure, without distinction, without binding — makes **no predictive claims**.

No Derivable Quantities

From "pure unrestrained information" no measurable quantity follows, no length scale, no coupling constant, no field equation. The postulate describes what is suspected *before* all structure — but without structure there is no derivation, and without derivation there is no prediction.

Compatible With Every Observation

Since nothing follows from the postulate, it cannot be refuted by anything. Every observation — whether it supports FFGFT, string theory, MOND, or standard cosmology — is compatible with "pure unrestrained information", because the postulate excludes no observation.

> **Important:** [Not Falsifiable = Not Physical] A theory compatible with every observation makes no physical claims. The postulate of "pure information" is therefore not a physical theory — it is a **metaphysical framing**. This is not inherently worthless, but it is a fundamentally different claim from that of FFGFT.

No Occlusion Mechanism Derivable

Atticus's own theory of "O-bits" and sequential adhesion (occlusion) makes interesting structural claims. But these do not follow from the postulate of "pure information" — they are introduced *additionally*, as a separate rule. This rule is itself already structure. The postulate of the primitive does not carry the burden of occlusion — it is imported from outside.

.4 The Philosophy-of-Science Distinction

FFGFT and Atticus's approach differ not only in the choice of primitive (T^4 vs. "pure information"), but in a deeper methodological point:

Criterion	FFGFT	"Pure information" (Myser)
Primitive	$T^4 + \xi$ (geometric, posited)	"pure unrestrained information" (posited)
Operational	Yes — derivations, field equations, measurement	No — no derivation possible
Falsifiable	Yes — L_0, D_f, singularities, α	No — no observation excluded
Predictions	Concrete, numerical	None
Connectable	To Standard Model, GR, QFT	Not directly
Type of claim	Physical theory	Metaphysical framing

[The Decisive Distinction] **FFGFT is a physical theory with empirical content.** The postulate of "pure information" is a metaphysical speculation without derivation power. The decisive difference lies not only in the choice of axiom, but in the *consequences* of that choice: what follows from the primitive? how much follows? and can what follows be refuted by observation?

.5 Falsifiability and Epistemic Openness

This distinction is not an attack on metaphysical framings. Metaphysical questions — why is there something rather than nothing? what is the essence of information? what lies before all structure? — are legitimate and important questions. They are simply not decidable by physical experiments.

FFGFT provides no answer to these questions. It leaves "why T^4?" explicitly open (cf. Dok. 248, Section 8). This is structural honesty: FFGFT describes *from within*, with the means available to an inside description.

What distinguishes FFGFT from metaphysical speculation is not the absence of an answer to the ultimate question — but the presence of concrete, refutable claims about the world we see.

FFGFT is falsifiable but epistemically modest. It makes no claims about what lies before T^4. It makes precise claims about what follows from $T^4 + \xi$ — and these claims can be refuted by observation. That is the claim of a physical theory, nothing more and nothing less.

Summary

Claim	Position
FFGFT makes no predictive claims	False — L_0, D_f, singularities, α are predictions

FFGFT is not falsifiable	False — each prediction is in principle refutable
“Pure information” is a physical theory	No — metaphysical framing without derivation power
“Pure information” is falsifiable	No — no observation is excluded
Both approaches are scientifically equivalent	No — only FFGFT is a physical theory in Popper’s sense
Metaphysical framings are worthless	No — they are legitimate, but of a different kind
FFGFT answers “why T^4?”	No — this question remains explicitly open (cf. Dok. 248)